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Correspondence to: Christine J. Ko, MD, Department of Dermatology, Yale University School of Medicine, 15 York St, LMP 5031, New Haven, CT 06510

E-mail: Christine.ko@yale.edu

Conflicts of interest

None disclosed.

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A stepwise approach for the management of primary erythromelalgia: A prospective single-arm study



To the Editor: Primary erythromelalgia (PEM) is a rare and highly disabling genetic disease characterized by severe burning pain and erythema of the extremities. PEM is caused by gain-of-function mutations in *SCN9A*, leading to the overactivation of sodium channels $Na_v1.7$.^{1,2} Its management remains challenging, with no consensus-based clinical guidelines. Sodium channel blockers such as mexiletine and carbamazepine have been used anecdotally to relieve pain in patients with PEM.³ Rizatriptan is a 5-HT_{1B/1D} receptor agonist that effectively controls acute migraine⁴ and neuropathic pain.⁵ The rapid onset of action (within 30 minutes) of rizatriptan⁴ might help alleviate acute pain attacks in patients with PEM; however, it has not been used previously in

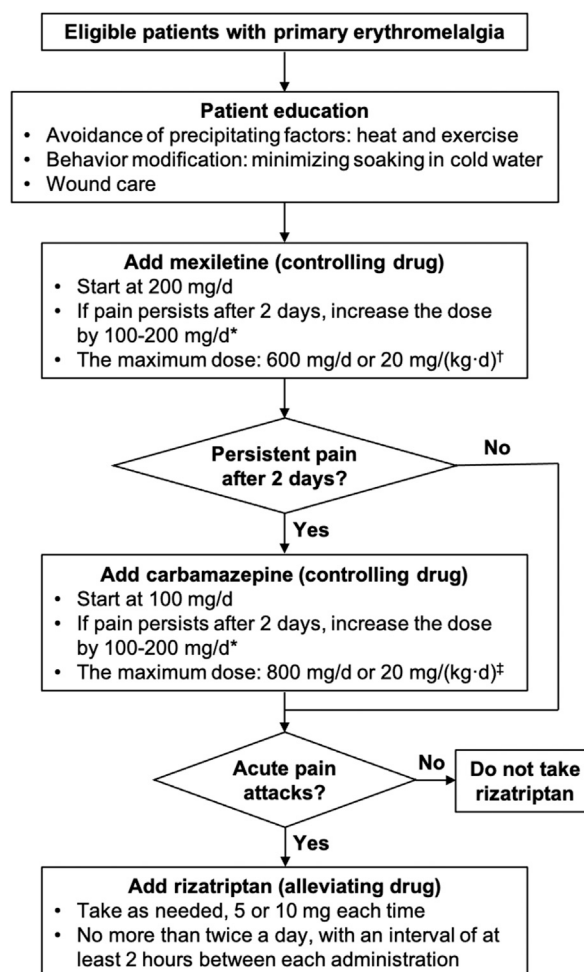


Fig 1. A stepwise approach for the management of primary erythromelalgia. The drug dosage was adjusted based on the patient's symptoms and BW. The total daily dose was taken in 2 or 3 divided doses. *Mexiletine and carbamazepine were uptitrated until the following: (1) a satisfactory pain control was achieved, (2) further dose increases failed to improve pain management or (3) the maximum dose was reached. If no response to mexiletine was observed, carbamazepine monotherapy could be given. †The maximum dose was 600 mg/d for patients with BW of ≥ 30 kg and 20 mg/(kg·d) for patients with BW of < 30 kg. ‡The maximum dose was 800 mg/d for patients with BW of ≥ 40 kg and 20 mg/(kg·d) for patients with BW of < 40 kg. BW, Body weight.

such cases. This prospective single-arm study investigated the efficacy and safety of a stepwise approach in treating PEM with these 3 drugs.

Patients clinically diagnosed with PEM and molecularly confirmed by the presence of pathogenic mutations in *SCN9A* were recruited. PEM was managed using a stepwise approach that consisted of patient education and sequential addition of oral drugs as needed (Fig 1). The drugs included 2

Table I. Clinical features, genotype, and treatment response of the 7 patients

Patient	Sex	Age, y	Body weight, kg	SCN9A	Family history	Onset age, y	Prior systemic treatments	Initial treatment plan				Treatment response		Average final NRS score* (range)
								Baseline NRS score	Mexiletine, mg/d	Carbamazepine, mg/d	Rizatriptan, mg each time	Pain improvement	Skin improvement	
1	F	14	42	c.406A>G	None	9	Gabapentin, aspirin	4	500	200	5 or 10	Complete	Complete	0.3 (0-1)
2	M	8	25	c.647T>C	None	5	Paracetamol, hydromorphone, prednisone, aspirin	5	400	100	None	Complete	Complete	1.8 (0-4)
3	F	12	41	c.2543T>C	None	3	Gabapentin, amitriptyline, propranolol, vitamin B6	6	600	200	5	Complete	Complete	0 (0)
4	F	8	32	c.2572C>T	None	4	None	5	300	None	10	Complete	Complete	0.5 (0-2)
5	M	7	29	c.2572C>T	None	6	Phenytoin, aspirin, traditional Chinese medicine	8	600	400	10	Major	Complete	2.8 (2-3)
6	M	12	48	c.3923C>T	Yes	7	None	4	300	200	None	Complete	Major	0 (0)
7	F	22	60	c.3947T>C	Yes	15	Aspirin, codeine, gabapentin, lumbar sympathectomy (in 2014)	5	400	400	10	Complete	Complete	0.1 (0-1)

NRS, Numerical rating scale.

*Average final NRS score = the average NRS score in the last 12 months of follow-up for every patient.

controlling drugs mexiletine and carbamazepine that were taken daily to maintain symptomatic control and an alleviating drug rizatriptan that was taken only as needed, aiming to quickly alleviate acute pain attacks.

Treatment response (pain and skin improvement) and adverse events were assessed by monthly follow-up for up to 24 months. The severity of pain was assessed using the numerical rating scale (NRS), ranging from 0 (no pain) to 10 (the worst pain possible). The detailed study design has been provided in Supplementary Methods (available via Mendeley at <https://data.mendeley.com/datasets/56tp93t52r/1>).

The study enrolled 7 unrelated patients with PEM (4 females, 3 males), with an average age of 11.9 years (range, 7-22 years). The baseline NRS score, initial treatment plan, and treatment response are summarized in Table I. Six (85.7%) patients had complete pain improvement (Supplementary Table I, available via Mendeley at <https://data.mendeley.com/datasets/56tp93t52r/1>). The average final NRS score of all 7 patients was significantly lower than the baseline score (mean, 5.3 vs 0.8; $P = .018$), indicating the maintenance of favorable treatment response over the follow-up period. Complete skin improvement was achieved in 6 (85.7%) patients (Supplementary Fig 1, available via Mendeley at <https://data.mendeley.com/datasets/56tp93t52r/1>). The beneficial effect of this regimen was also reflected by improvement in sleep quality and spontaneous remission of comorbidities (hypertension and recurrent diarrhea). Notably, 3 patients returned to school after treatment (Supplementary Results, available via Mendeley at <https://data.mendeley.com/datasets/56tp93t52r/1>). Two patients reported temporary mild dizziness after taking carbamazepine. No cardiovascular or serious adverse events were observed.

We propose a newly designed regimen incorporating patient education, treatment of underlying pathophysiology (mexiletine and carbamazepine), and symptomatic management of acute pain (rizatriptan). Despite some limitations, including a smaller sample size, uncertain safety profile, and relatively low NRS scores at baseline, this regimen might represent a reliable initial treatment plan for PEM, leading to significant long-term improvement with a favorable tolerability profile. It should be noted that the medications were used in a stepwise manner; this meant that not all 3 medications were required for each patient, and doses could be reduced or even discontinued after the pain reduced. Also, the regimen was trialed in a defined subset of patients with erythromelalgia (PEM with

SCN9A mutations) and might not apply to all patients.

Zijuan Wang, MD,^a Huijun Wang, PhD,^b
Mingyang Lee, MD, PhD,^a Mao-Ying Lin, MD,^c
and Zhimiao Lin, MD^{a,b,c}

From the Department of Dermatology, Peking University First Hospital, Beijing Key Laboratory of Molecular Diagnosis on Dermatoses, National Clinical Research Center for Skin and Immune Diseases, Beijing, China^a; Dermatology Hospital, Southern Medical University, Guangzhou, China^b; and Department of Dermatology, Xiamen Chang Gung Hospital, Xiamen, China.^c

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Correspondence and reprint requests to: Zhimiao Lin, MD, and Huijun Wang, PhD, Dermatology Hospital, Southern Medical University, No.2 Lujing St, Guangzhou 510091, China

E-mail: zhimiaolin@bjmu.edu.cn or drhuijunwang@pku.edu.cn

Conflicts of interest

None disclosed.

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Low-dose oral minoxidil for treatment of androgenetic alopecia and telogen effluvium in a pediatric population: A descriptive study



To the Editor: Topical minoxidil has been described for the treatment of children's hair disorders.^{1,2} Despite its safety, local irritant dermatitis and difficulties in its application may limit adherence.^{1,2} The objective of our study is to describe the use of low-dose oral minoxidil (LDM) for the treatment of hair disorders in a pediatric population in order to broaden therapeutic options in this age group.

We retrospectively reviewed medical records of patients <18 years who attended a trichology-specialized clinic in Spain for a period of 2 years (between March 2019 and March 2021). From among these patients, we selected those patients who were diagnosed with androgenetic alopecia or telogen effluvium and had received treatment with LDM. The epidemiologic and clinical features of the selected patients were collected.

In total, 45 patients with a mean age of 16 years (range, 10-17 years) were included in our study (Table 1). All the patients weighed >40 kg. Diagnosis was determined upon clinical and trichoscopic findings. Thirty-nine patients (87%) were diagnosed with androgenetic alopecia and 6 patients (13%) were diagnosed with telogen effluvium. LDM was used in monotherapy in 14 children (31%), combined with other topical treatments (topical minoxidil or topical finasteride) in 8 patients (18%), and used with other oral treatments (finasteride, dutasteride, bicalutamide or spironolactone) in 23 patients (51%). Minoxidil was prescribed and compounded at a mean daily dose of 1.24 mg in 23 patients, with lower doses in girls (mean dosage 0.63 mg) compared to boys (mean dosage 2.35 mg). Although dose election was personalized, we usually started with 0.5 mg daily, doubling it in each subsequent visit if more improvement was needed.

Forty patients completed at least 1 follow-up visit to assess LDM's effectiveness and safety. At the time of analysis, treatment duration was a mean of 9.8 months (range, 3-24), with most patients continuing with therapy. Clinical improvement in hair density was observed in 32 patients (80%) (Supplementary Fig 1, available via Mendeley at <https://data.mendeley.com/datasets/k8m7rby4bf/1>), with a hair loss stabilization in the remaining 8 patients (20%). Adverse effects were reported in 10 children (25%), most commonly facial hypertrichosis (18%), followed by shedding (2 patients) and